

Title:	----
Part Number:	Engine harness
Description:	



Rev.	Date	Author	Description

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Flag	Quantity	Note

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Id	Type	Manufacturer	Part Number	Description	Quantity	Usage
2	Connector		2 Cavity	Generic Connector	13	CKP, CMP, Coil 1, Coil 2, Coil 3, Coil 4, EOP / Neutral, IAT, Injector 1, Injector 2, Injector 3, Injector 4, O2
3	Connector		3 Cavity	Generic Connector	5	ECT, Knock, MAP, TPS, VS Sensor
5	Wire		20 AWG Blue	Generic Wire	≥ 41 ft 5 in	W1, W2, W3, W4, W5, W6, W7, W8, W9
6	Wire		20 AWG Black	Generic Wire	≥ 32 ft 6 in	W10, W11, W12, W13, W16, W17, W18, W19, W50
7	Wire		20 AWG Orange	Generic Wire	≥ 7 ft 11 in	W14, W15, W20, W49
8	Wire		16 AWG Red	Generic Wire	≥ 412 in	W21, W22, W23, W24, W41, W42, W43, W44, W45, W46, W52
9	Wire		16 AWG Black	Generic Wire	≥ 334 in	W25, W26, W27, W28, W35, W36, W37, W38, W47
10	Wire		16 AWG Yellow	Generic Wire	0 in	W29, W30, W31, W32, W33, W34
12	Wire		UL1015 12 AWG Black	UL 1015 rated 12 AWG Black PVC Hook-up wire	≥ 2.42 ft	W39, W40
13	Wire		16 AWG Orange	Generic Wire	2.5 ft	W48
14	Wire		UL1015 12 AWG Red	UL 1015 rated 12 AWG Red PVC Hook-up wire		W51, W53
17	Connector	TE Connectivity	DTP06-2S	Automotive Connector Housing, IP68, DTP Series, Plug, 2 Ways, AMP DTP Series Socket Contacts	2	Fan, Fuel Pump
18	Connector	TE Connectivity	DRC26-40SA	DEUTSCH DRC Series 40 Position 4 Row 4.42 mm Pitch Black Housings Plug Connector	1	40-pin
19	Connector	TE Connectivity	DTP04-4P	Connector, Hybrid, Male Housing, Gray, 2.5MM, 4 Position, Centerline, Dtp Series	1	Power
20	Wire		18 AWG Black	Generic Wire	4.75 ft	W54

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From	To	Conductor	Conductor PN	Twisted With	Size	Length	From Contact PN	Notes
CKP.1	40-pin.7	 W1.Blue	20 AWG Blue		20 AWG	4.92 ft		CKP (+)
CKP.2	40-pin.6	 W11.Black	20 AWG Black		20 AWG	4.92 ft		CKP (-)
CMP.1	40-pin.9	 W3.Blue	20 AWG Blue		20 AWG	52 in		CMP (+)
CMP.2	40-pin.8	 W10.Black	20 AWG Black		20 AWG	52 in		CMP (-)
Coil 1.1	S1	 W43.Red	16 AWG Red		16 AWG	7 in		Coil 1 Shutdown
Coil 1.2	40-pin.22	 W35.Black	16 AWG Black		16 AWG	51 in		Coil 1 Ground (Module 1)
Coil 2.1	S2	 W44.Red	16 AWG Red		16 AWG	11 in		Coil 2 Shutdown
Coil 2.2	40-pin.13	 W36.Black	16 AWG Black		16 AWG	55 in		Coil 2 Ground (Module 1)
Coil 3.1	S2	 W45.Red	16 AWG Red		16 AWG	15 in		Coil 3 Shutdown
Coil 3.2	40-pin.14	 W37.Black	16 AWG Black		16 AWG	59 in		Coil 3 Ground (Module 2)
Coil 4.1	S1	 W46.Red	16 AWG Red		16 AWG	19 in		Coil 4 Shutdown
Coil 4.2	40-pin.15	 W38.Black	16 AWG Black		16 AWG	63 in		Coil 4 Ground (Module 2)
CoilPower	Power.1	 W52.Red	16 AWG Red		16 AWG			Main Coil Power
ECT.1	40-pin.17	 W6.Blue	20 AWG Blue		20 AWG	5.75 ft		ECT (Data)
ECT.2	Sensor Ground	 W17.Black	20 AWG Black		20 AWG	39 in		ECT (Ground)
EOP / Neutral.1	40-pin.1	 W12.Black	20 AWG Black		20 AWG	4.92 ft		Neutral Switch
EOP / Neutral.2	40-pin.4	 W19.Black	20 AWG Black		20 AWG	4.92 ft		EOP Switch
Fan.1	Power.3	 W53.Red	UL1015 12 AWG Red		12 AWG			Fuel Pump 12V
Fan.2	Engine Ground	 W40.Black	UL1015 12 AWG Black		12 AWG			
Fuel Pump.1	Power.2	 W51.Red	UL1015 12 AWG Red		12 AWG			Fuel Pump 12 V
Fuel Pump.2	Engine Ground	 W39.Black	UL1015 12 AWG Black		12 AWG	≥ 2.42 ft		Fuel Pump Ground
IAT.1	40-pin.2	 W4.Blue	20 AWG Blue		20 AWG	6.25 ft		IAT (Data)
IAT.2	Sensor Ground	 W13.Black	20 AWG Black		20 AWG	3.75 ft		IAT (Ground)
Injector 1.1	40-pin.26	 W21.Red	16 AWG Red		16 AWG	65 in		Injector 1 (12V)
Injector 1.2	InjectorSplice1	 W25.Black	16 AWG Black		16 AWG	≥ 13 in		Injector 1 (ECU)
Injector 2.1	40-pin.27	 W22.Red	16 AWG Red		16 AWG	69 in		Injector 2 (12V)
Injector 2.2	InjectorSplice2	 W26.Black	16 AWG Black		16 AWG	≥ 17 in		Injector 2 (ECU)
Injector 3.1	40-pin.28	 W23.Red	16 AWG Red		16 AWG	73 in		Injector 3 (12V)
Injector 3.2	InjectorSplice2	 W27.Black	16 AWG Black		16 AWG	≥ 21 in		Injector 3 (ECU)
Injector 4.1	40-pin.29	 W24.Red	16 AWG Red		16 AWG	77 in		Injector 4 (12V)
Injector 4.2	InjectorSplice1	 W28.Black	16 AWG Black		16 AWG	≥ 25 in		Injector 4 (ECU)
InjectorSplice1	40-pin.34					2.5 ft		
InjectorSplice2	40-pin.35					2.5 ft		
Knock.1	40-pin.38	 W2.Blue	20 AWG Blue		20 AWG	4.75 ft		Knock (Data)
Knock.2	Sensor Ground	 W18.Black	20 AWG Black		20 AWG	27 in		Knock (Ground)
Knock.3	40-pin.40	 W54.Black	18 AWG Black		18 AWG	4.75 ft		Knock Shielding
MAP.1	40-pin.16	 W7.Blue	20 AWG Blue		20 AWG	6.25 ft		MAP (Data)
MAP.2	Sensor 5V	 W15.Orange	20 AWG Orange		20 AWG	3.75 ft		MAP (5V)
MAP.3	Sensor Ground	 W16.Black	20 AWG Black		20 AWG	3.75 ft		MAP (Ground)
O2.1	40-pin.3	 W5.Blue	20 AWG Blue		20 AWG			O2 (Data)
S1	CoilPower	 W41.Red	16 AWG Red		16 AWG	3.17 ft		Coil 1 and 4 Shutdown

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S2	CoilPower	 W42.Red	16 AWG Red		16 AWG	3.17 ft		Coil 2 and 3 Shutdown
Sensor 5V	40-pin.36	 W48.Orange	16 AWG Orange		16 AWG	2.5 ft		Sensor 5V
Sensor Ground	40-pin.10	 W47.Black	16 AWG Black		16 AWG	2.5 ft		Sensor Ground
TPS.1	40-pin.11	 W8.Blue	20 AWG Blue		20 AWG	6.25 ft		TPS (Data)
TPS.2	Sensor 5V	 W20.Orange	20 AWG Orange		20 AWG	3.75 ft		TPS (5V)
VS Sensor.1	Sensor 5V	 W49.Orange	20 AWG Orange		20 AWG	≥ 5 in		VS Speed 5v
VS Sensor.2	40-pin.18	 W9.Blue	20 AWG Blue		20 AWG	≥ 2.92 ft		VS Speed Data
VS Sensor.3	Sensor Ground	 W50.Black	20 AWG Black		20 AWG	≥ 5 in		VS Speed Ground

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#	Type	Reference Designator	Manufacturer	Part Number	Color	Size	From Strip	To Strip	Length	Twisted With	Notes
1	Wire	W1		20 AWG Blue	 Blue	20 AWG			4.92 ft		CKP (+)
2	Wire	W2		20 AWG Blue	 Blue	20 AWG			4.75 ft		Knock (Data)
3	Wire	W3		20 AWG Blue	 Blue	20 AWG			52 in		CMP (+)
4	Wire	W4		20 AWG Blue	 Blue	20 AWG			6.25 ft		IAT (Data)
5	Wire	W5		20 AWG Blue	 Blue	20 AWG					O2 (Data)
6	Wire	W6		20 AWG Blue	 Blue	20 AWG			5.75 ft		ECT (Data)
7	Wire	W7		20 AWG Blue	 Blue	20 AWG			6.25 ft		MAP (Data)
8	Wire	W8		20 AWG Blue	 Blue	20 AWG			6.25 ft		TPS (Data)
9	Wire	W9		20 AWG Blue	 Blue	20 AWG			≥ 2.92 ft		VS Speed Data
10	Wire	W10		20 AWG Black	 Black	20 AWG			52 in		CMP (-)
11	Wire	W11		20 AWG Black	 Black	20 AWG			4.92 ft		CKP (-)
12	Wire	W12		20 AWG Black	 Black	20 AWG			4.92 ft		Neutral Switch
13	Wire	W13		20 AWG Black	 Black	20 AWG			3.75 ft		IAT (Ground)
14	Wire	W15		20 AWG Orange	 Orange	20 AWG			3.75 ft		MAP (5V)
15	Wire	W16		20 AWG Black	 Black	20 AWG			3.75 ft		MAP (Ground)
16	Wire	W17		20 AWG Black	 Black	20 AWG			39 in		ECT (Ground)
17	Wire	W18		20 AWG Black	 Black	20 AWG			27 in		Knock (Ground)
18	Wire	W19		20 AWG Black	 Black	20 AWG			4.92 ft		EOP Switch
19	Wire	W20		20 AWG Orange	 Orange	20 AWG			3.75 ft		TPS (5V)
20	Wire	W21		16 AWG Red	 Red	16 AWG			65 in		Injector 1 (12V)
21	Wire	W22		16 AWG Red	 Red	16 AWG			69 in		Injector 2 (12V)
22	Wire	W23		16 AWG Red	 Red	16 AWG			73 in		Injector 3 (12V)
23	Wire	W24		16 AWG Red	 Red	16 AWG			77 in		Injector 4 (12V)
24	Wire	W25		16 AWG Black	 Black	16 AWG			≥ 13 in		Injector 1 (ECU)
25	Wire	W26		16 AWG Black	 Black	16 AWG			≥ 17 in		Injector 2 (ECU)
26	Wire	W27		16 AWG Black	 Black	16 AWG			≥ 21 in		Injector 3 (ECU)
27	Wire	W28		16 AWG Black	 Black	16 AWG			≥ 25 in		Injector 4 (ECU)
28	Wire	W35		16 AWG Black	 Black	16 AWG			51 in		Coil 1 Ground (Module 1)
29	Wire	W36		16 AWG Black	 Black	16 AWG			55 in		Coil 2 Ground (Module 1)
30	Wire	W37		16 AWG Black	 Black	16 AWG			59 in		Coil 3 Ground (Module 2)
31	Wire	W38		16 AWG Black	 Black	16 AWG			63 in		Coil 4 Ground (Module 2)
32	Wire	W39		UL1015 12 AWG Black	 Black	12 AWG			≥ 2.42 ft		Fuel Pump Ground
33	Wire	W40		UL1015 12 AWG Black	 Black	12 AWG					
34	Wire	W41		16 AWG Red	 Red	16 AWG			3.17 ft		Coil 1 and 4 Shutdown
35	Wire	W42		16 AWG Red	 Red	16 AWG			3.17 ft		Coil 2 and 3 Shutdown
36	Wire	W43		16 AWG Red	 Red	16 AWG			7 in		Coil 1 Shutdown
37	Wire	W44		16 AWG Red	 Red	16 AWG			11 in		Coil 2 Shutdown
38	Wire	W45		16 AWG Red	 Red	16 AWG			15 in		Coil 3 Shutdown
39	Wire	W46		16 AWG Red	 Red	16 AWG			19 in		Coil 4 Shutdown
40	Wire	W47		16 AWG Black	 Black	16 AWG			2.5 ft		Sensor Ground
41	Wire	W48		16 AWG Orange	 Orange	16 AWG			2.5 ft		Sensor 5V

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#	Type	Reference Designator	Manufacturer	Part Number	Color	Size	From Strip	To Strip	Length	Twisted With	Notes
42	Wire	W49		20 AWG Orange	 Orange	20 AWG			≥ 5 in		VS Speed 5v
43	Wire	W50		20 AWG Black	 Black	20 AWG			≥ 5 in		VS Speed Ground
44	Wire	W51		UL1015 12 AWG Red	 Red	12 AWG					Fuel Pump 12 V
45	Wire	W52		16 AWG Red	 Red	16 AWG					Main Coil Power
46	Wire	W53		UL1015 12 AWG Red	 Red	12 AWG					Fuel Pump 12V
47	Wire	W54		18 AWG Black	 Black	18 AWG			4.75 ft		Knock Shielding
48		InjectorSplice1 - 40-pin.34							2.5 ft		
49		InjectorSplice2 - 40-pin.35							2.5 ft		

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#	Manufacturer	Part Number	Description	Label Text